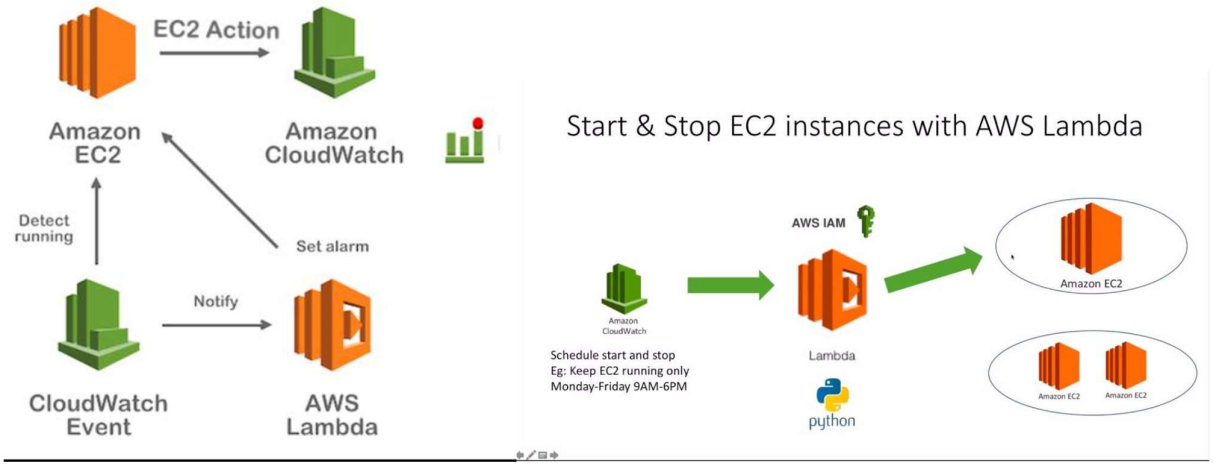
Name : Vivek Chaudhari

Project -

EcoSave : Automated AWS cost reduction using Lambda



Amazon EC2 → Launch EC2 instance for testing

AWS Lambda → Create function to automate logic

Amazon CloudWatch → Monitor EC2 CPU usage

Lambda → Fetch all running EC2 instances

Lambda → Check CPU utilization of each instance

Decision → If CPU < 10% (idle)

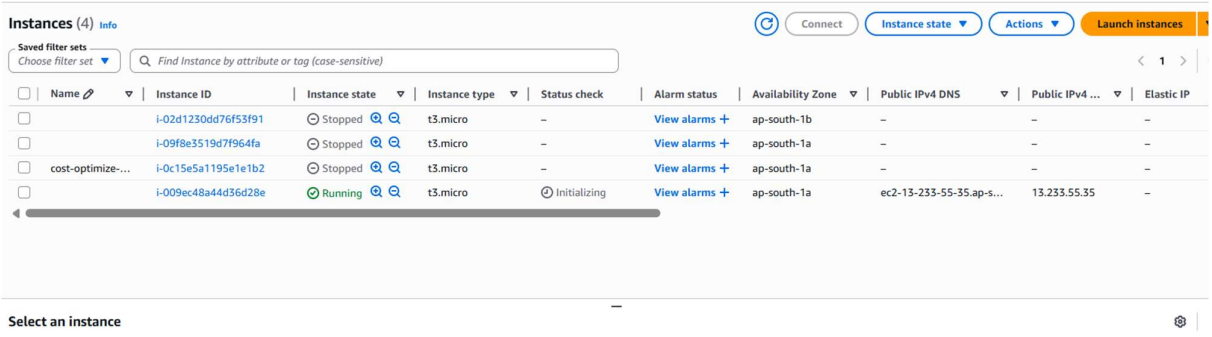
Action → Stop EC2 instance automatically

CloudWatch → Trigger Lambda on schedule

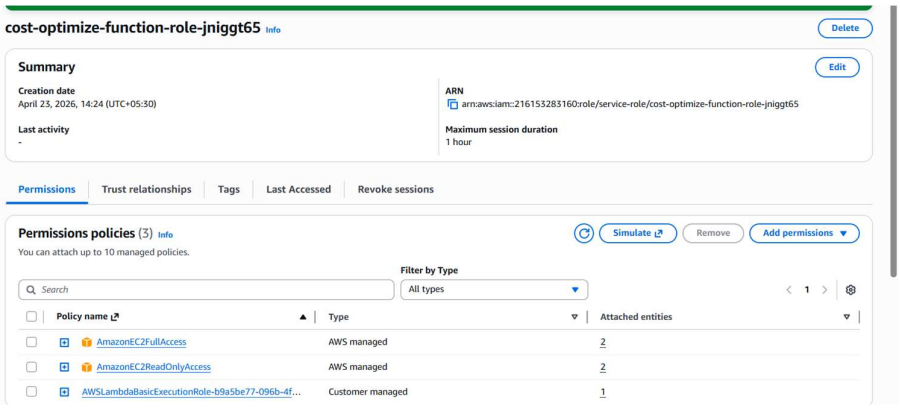
Output → Idle instances stopped successfully

Result → AWS cost optimized 💰

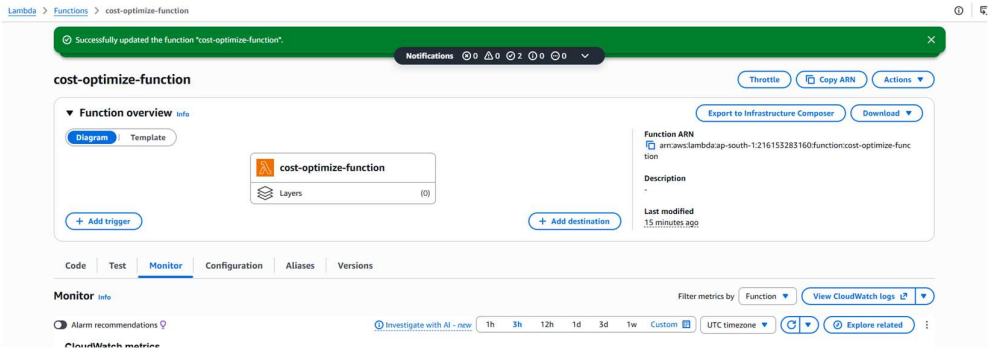
Step 1 : Create EC2 Instance :



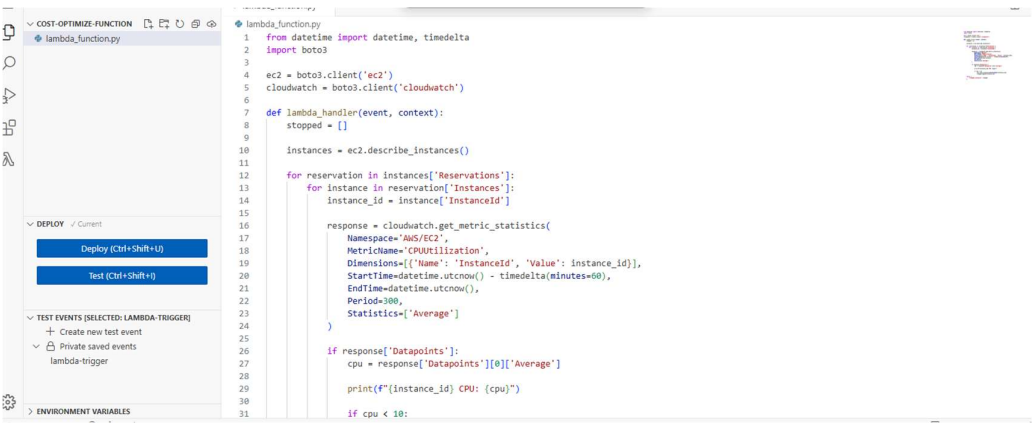
Step 2 : Cerate IAM Role and Attach Policy ;



Step 3 : Create Lambda Function and attach role policy

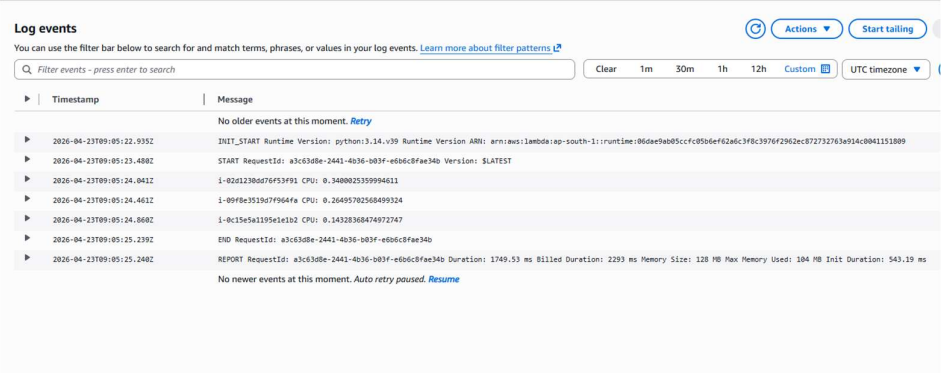


Step 4 : Code Optimize :




```
        Period=300,
        Statistics=['Average']
    )

    if response['Datapoints']:
        cpu = response['Datapoints'][0]['Average']

        print(f"{instance_id} CPU: {cpu}")

        if cpu < 10:
            ec2.stop_instances(InstanceIds=[instance_id])
            stopped.append(instance_id)

return {
    "stopped_instances": stopped
}
```

Step 5 : Final Output :



Summary of Project :

The **EcoSave** project is an automated cost optimization system built on **Amazon Web Services** that helps reduce unnecessary cloud expenses by intelligently managing compute resources. It uses **AWS Lambda** to execute automation logic, **Amazon CloudWatch** to track CPU utilization, and **Amazon EC2** instances as the resources being optimized. The system periodically checks the CPU usage of running instances, and if any instance is found to be underutilized (below a defined threshold), it automatically stops it. This eliminates manual monitoring, improves resource efficiency, and significantly reduces AWS costs, making it a practical and scalable solution for cloud cost management.